

TOA: Theory of Machines.

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a and b are two real numbers.

can you check if the two are same?

if $(a = b)$?

NO.

practically yes. But as a technical you can't.

What is a computer?

FINITE AUTOMATA

→ Simplest models of computation

→ we build complex ones using this.

↳ computational model: simplest.

What is a computational model?

→ compute yes/no questions.

ans could only be binary. Can't use to sort no, find max etc. so very very basic.

→ Sometimes saying yes/no as reject/accept, true/false. → So basically a binary answer.

→ the questions are also only binary.

String:

in Java characters joined together is a string which is an abstract data type

language:

is a set of strings.

in computers we have formal languages (not natural languages).

in TDA, if you have any set that contains words is a language.

$\Sigma \rightarrow$ alphabet

set that contains the building blocks in the lang
(Sigma.

$\Sigma = \{a, b, c\}$ Only 3 alphabets in this set - however, we can make infinitely many words.

$\Sigma = \{a\}$ single character in alphabet - sigma can be used to make infinitely many words

Σ is the alphabet-

A is a set that contains

$$A = \{a, aa, ba, ca, aaaa, abca, \dots\}$$

(The words in A end in a . (property given))

Is A a subset of all possible words over Σ ? Yes. This set A is a language

Size of A is infinite

We can create infinitely many languages

eg: $B = \{w \mid \text{length of } w \text{ is even}\}$
 → infinite set (infinitely many words)
 → every word in B can be formed from set Σ with even length.

Any subset of words over any finite alphabet is a language.
 Σ (can be finite, can be infinite)

→ Any set is a subset of itself so
 Σ or itself is a language.
 → ~~it~~ itself is a language.

$C = \{aa, bb, cc\} \rightarrow$ also a language

Σ should be finite, and has a minimum of 1 character. Σ is Fixed (can't ever be changed).

once you've started your argument, you can't change elements of Σ .

we can combine elements of Σ to form a word.

A set can't have repeated elements

we combine alphabets to form word
A subset of finite/infinite amount of words are called language

Empty String:

→ in Java ""

→ in TOA ϵ → has length 0.

Epsilon ϵ is a string of length zero.

↳ This means there's always a length of string. length is number of characters.

in string.

This course answers one kind of question.. Given a string w and a language L . does w belong to L ?

↳ All that we will be doing for the entire course

Set of words or collection of words
→ language

alphabets are single characters.
words can be single character as well.

Membership Query:

→ type of computational model.
→ called membership query as asks if something is a member or not.

x^k , k can not be negative

↳ k copies of x .

eg $x = 01$

$$x^2 = 0101 = xx$$

$k=0$. zero copies of
 $x =$ empty
string

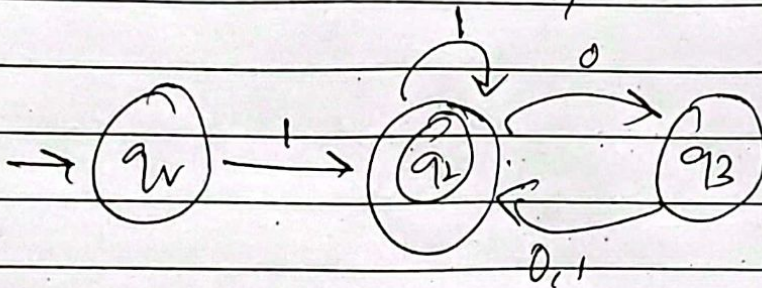
any string to power 0 is an empty string.

○ → state.

⊙ → double circle. Accepting state

→ arrow coming to a circle from nowhere. the circle it goes to is the initial starting state when machine is turned on.

finite Automata, means finite machine.



we are passing a string q for language L .
it checks if q does or does not belong to L .

The ans is yes/no.

reads the entire string left to right, one character at a time.

when machine turns ON, it starts in state q_0 .

the empty arrow coming from nowhere identifies ~~as~~ start state.

our language contains 2 characters 0 or 1.

if it reads 0, stay in q_1 . if it reads 1 it moves to q_2 .

if machine is in state q_2 , if it reads 1 do nothing - stay there.

machine stops when last character scanned. we will find which state it ended in. it can end many of q_1, q_2 or q_3 .

ending in $q_1/q_3 \rightarrow$ rejected.
 " " $q_2 \rightarrow$ accepted.

All that is accepted is in the language

possible problems.

lang to machine.
 machine to lang.

5 tuple means there are 5 things that make a finite automata:

$$Q \rightarrow q_1, q_2, q_3$$

$$\Sigma \rightarrow 0, 1$$

$$q_0 \rightarrow q_1 \quad \text{only 1 starting state}$$

$$F \rightarrow q_2 \quad \text{possible to have more than 1}$$

$$\delta \rightarrow \text{transition from } q_1 \text{ to } q_2 \text{ to } q_3 \text{ etc.}$$

delta. when input is read.

Fluent in set theory.